BRAC UNIVERSITY

Inspiring Excellence

Assignment : 04

Course Code : CSE320

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Chapter - 06

FDM:

1/ Here, we have 5 channels. so, guard band should be = (5-1) = 4.

As each channel has 20 KHz, so, channels band width = 5×20 = 100 KHz

guard band's bandwidth = (4×5) = 20 KHz.

so, Output channel band width = (100+20) = 120 KHz.

Ans

2/ Here, we have 5 channels, so, guard bands should be, (5-1) = 4. And bandwidth of the guard bands = (4×5) = 20 KHz.

Output band width is 100 KHz.

total band width of the 5 channel = (100-20) = 80 KHz

Max band width of each channel = $\frac{80}{5}$ = 16 KHz

Ans

3| Here, Output bandwidth = 100KHz.

5 channels with 20KHz bandwidths.

So, channel bandwidth = 5×20 = 100KHz

So, Guard band bandwidth = (100-100) = 0 KHz

So, there is no room for any guard bands.

4| From our given picture.

in 1st Fdm,

channels = 12, bandwidth of each = 4 KHz

guard bands = 12-1 = 11, bandwidth of each guard band = 1 KHz

band width of output of a group

= 12×4 + 11×1 = 59 KHz

~~channels in a group = 4×12 = 48~~

Now, in super group we have, 5 groups with guard band of 5KHz.

Guard band required = (5-1) = 4, Voice

~~channels in super group = 48×5 = 240~~

Bandwidth of a super group $= 59 \times 5 + 4 \times 5$
$= 315$ KHz

Voice channels in super group $= 5 \times 12 = 60$

Now, in Master group there are 10 supergroups

so, guard band needed $= (10-1) = 9$, each with 10 KHz.

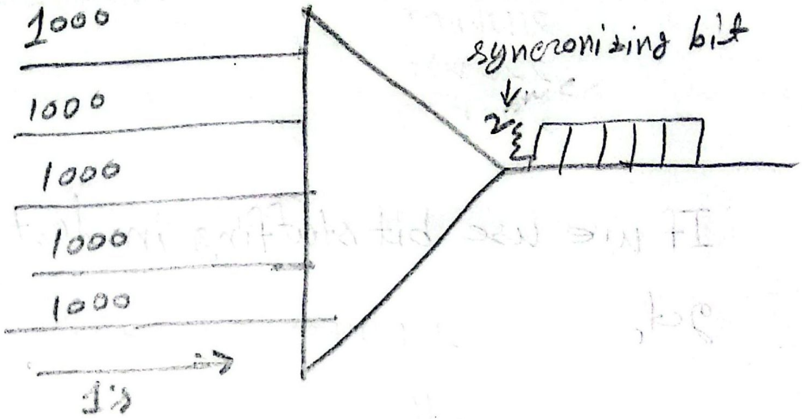
~~Chan~~ voice channels in supergroup $= (60 \times 10) = 600$.
(ans)

Output bandwidth of super group $= 315 \times 10 + 9 \times 10$
$= 3240$ KHz
(ans)

600 voice channels can be multiplexed in the supergroup and the required bandwidth will be 3240 KHz.

TDM:

1) Channels = 5,



ⓐ Inputrate,

= 1000 bps

$$\text{Input duration} = \frac{1}{\text{Inputrate}} = \frac{1}{1000} = 1\ ms$$

b) As it can multiplex 5 bits at a time, so, slotsize = 5

$$\text{frame rate} = \frac{\text{Inputrate}}{\text{slotsize}} = \frac{1000}{5} = 200\ fps$$

$$\text{frame duration} = \frac{1}{\text{framerate}} = \frac{1}{200} = 5\ ms$$

c) As we have 2 synchronizing bits, our frame size will be, $5 \times 5 + 2 = 27$

So, Output rate = frame size × frame rate

$= 27 \times 200$

$= 5400$ bps

$$\text{Output duration} = \frac{1}{5400} = 0.185\ ms$$

Ans to ques - 02

We have 5 channels, 1000 page in 5 second.

so, 1 second for $= \frac{1000}{5} = 200$ pages.

a) As each page has 20 lines with 40 characters,

our ~~output~~ input rate $= 200 \times 20 \times 40 \times 8$

$= 1280 \times 10^3$ bps

so, input duration $= \frac{1}{1280 \times 10^3} = 7.812 \times 10^{-7}$ s

b). As 5 lines at a time to be multiplexed,

in bits $= (5 \times 8) = 40$ bits

slot size $= 40$,

Frame rate $= \frac{1280 \times 10^3}{40} = 32 \times 10^3$ fps

frame duration $= \frac{1}{32 \times 10^3} = 3.125 \times 10^{-5}$ s

c) We have 5 synchronization bits, so,

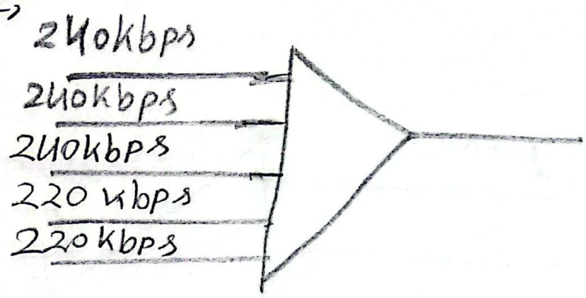
frame size $= 5 \times 40 + 5 = 205$ bits

output rate $= 205 \times 32 \times 10^3 = 6560 \times 10^3$ bps

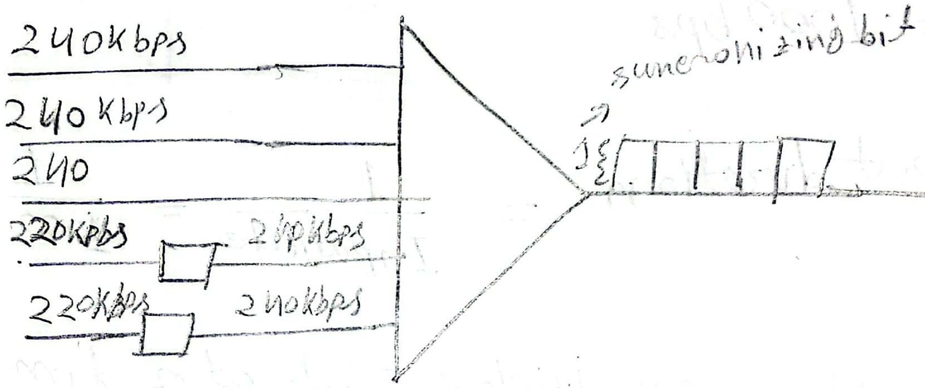
$= 6560$ Kbps

$$\text{output duration} = \frac{1}{6560 \times 10^{3}} = 1.52 \times 10^{-7} s$$

3] Here,



If we use bit stuffing in last two channels we get,



a) So, Input rate here is 240kbps.

Input Duration $= \frac{1}{240 \times 10^3} = 4.167 \times 10^{-6}$ s

b) Assuming 1 bit from each channel,

so, slot size = 1 bit

Frame rate $= \frac{240 \times 10^3}{1} =$ ~~240fps~~ 240×10^3 fps

frame duration $= \frac{1}{240 \times 10^3} = 4.167 \times 10^{-6}$ s

As there is one synchronizing bit, our frame size $= (5 \times 1 + 1) = 6$ bits

c) Output rate $= 6 \times 240 \times 10^3$

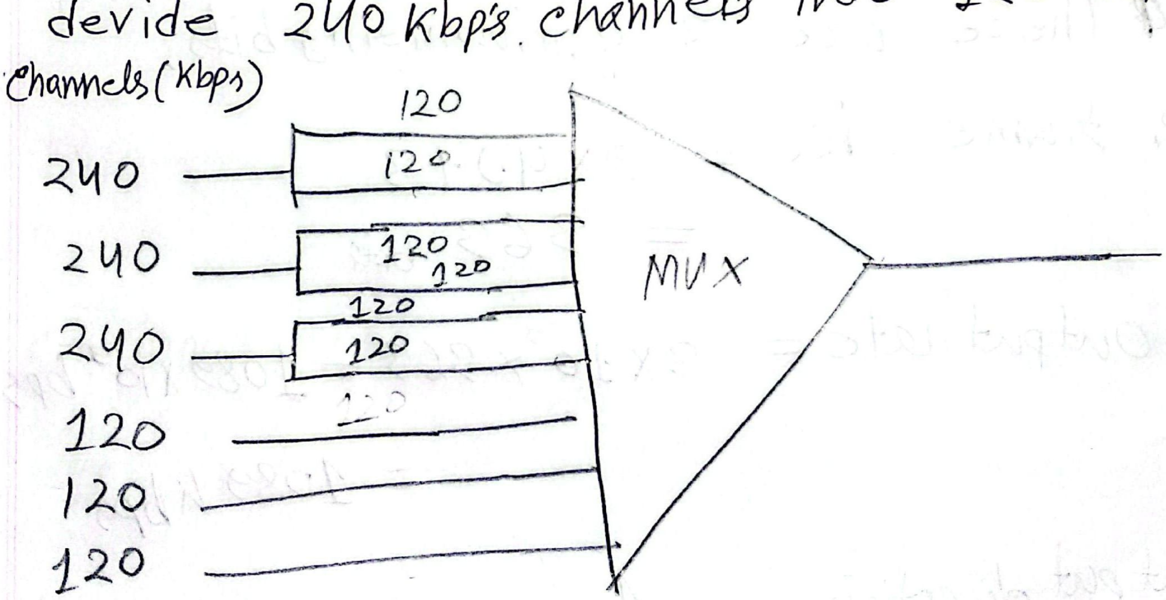
$= 1440 \times 10^3$ bps

$\Rightarrow 1440$ kbps

Output duration $= \frac{1}{1440 \times 10^3} = 6.94 \times 10^{-7}$ s

Ans to ques-04

a) Here we can use multilevel multiplexing to devide 240 kbps channels into 120 kbps channells.



Here, input rate is 120 kbps for all the channels,

Input duration $= \frac{1}{120 \times 10^3} = 8.33 \times 10^{-6}$ s b/ Here we got 9 channels after multiplexing.

So, We can take 5 character from each channel, ~~total~~ bits per channel = $5 \times 8 = 40$ bits.

Now, So slot size = 40 bits

frame rate = $\frac{120 \times 10^3}{40} = 3 \times 10^3$ fps

frame duration = $\frac{1}{3 \times 10^3} = 3.33 \times 10^{-4}$ s

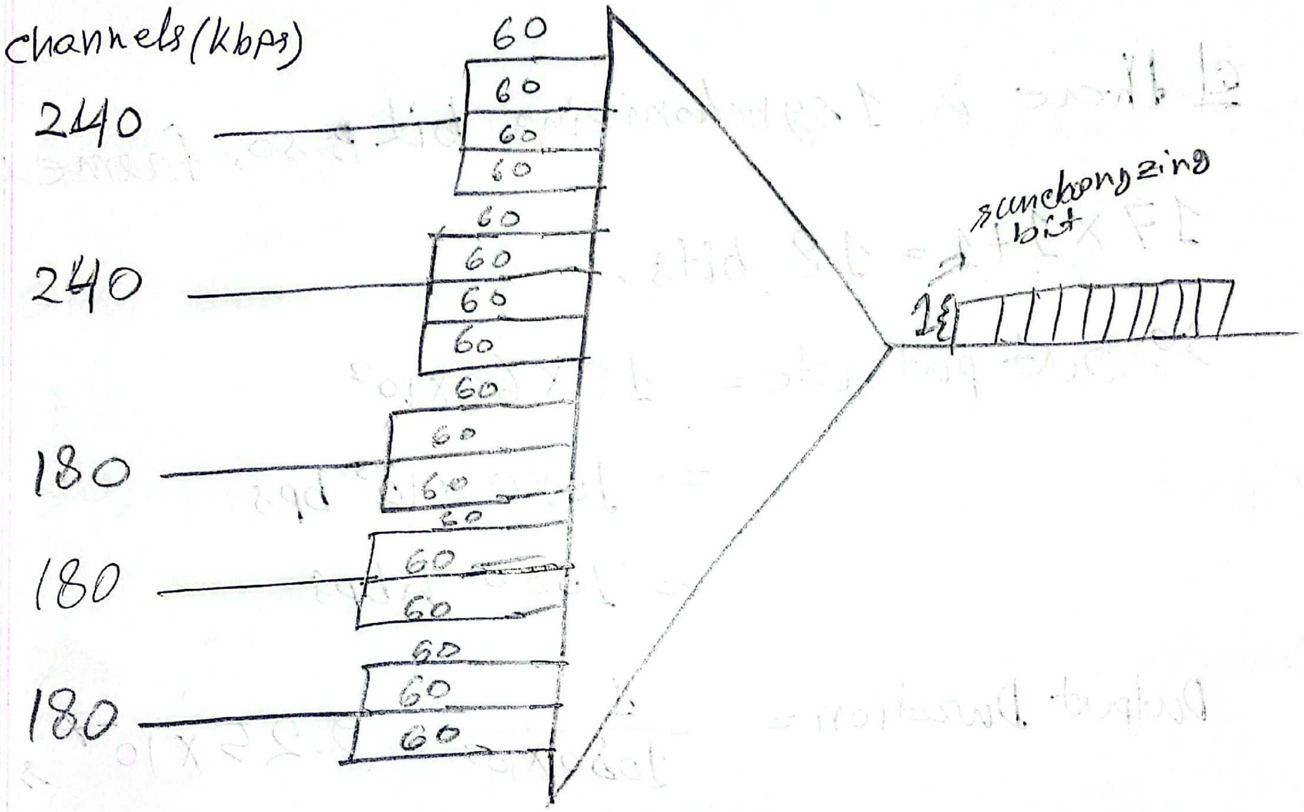
c/ There are 3 synchronizing bits,

so, frame size = $9 \times 40 + 3$
$= 363$ bits

So, output rate = $3 \times 10^3 \times 363 = 1089 \times 10^3$ bps
$= 1089$ Kbps

Output duration = $\frac{1}{1089 \times 10^3} = 9.18 \times 10^{-7}$ s

5| a) Here we can use multileveling to convert the 240 Kbps and 180Kbps channels into 60 Kbps channels.



So, input rate for the channels = 60 kbps,

$$\text{Input duration} = \frac{1}{60\times10^3} = 1.67\times10^{-5}\ s$$

b) Assuming 1 bit per channel, the slot size = 1

So, frame rate $= \frac{60\times10^3}{1} = 60\times10^3$ fps ~~6.67×10^3 fps~~

$$\text{frame Duration} = \frac{1}{\cancel{6.67\times10^3}\ 60\times10^3} = \cancel{1.5\times10^{-4}}\ 1.67\times10^{-5}\ s$$

d) we got a total $(4+4+3+3+3) = 17$ channels after multiplexing.

e) there is 1 synchronizing bit; so, framesize,

$17 \times 1 + 1 = 18$ bits.

so, output rate $= 18 \times 60 \times 10^3$

$= 1080 \times 10^3$ bps

$= 1080$ Kbps

Output Duration $= \frac{1}{1080 \times 10^3} = 9.25 \times 10^{-7}$ s

Chapter- 10

1)

Dataword	Codeword
00	11100000
01	00011101
10	10101010
11	01010111

Minimum hamnind distances.

(11100000, 00011101) = 7

(11100000, 10101010) = 3

(11100000, 01010111) = 6

(00011101, 10101010) = 6

(00011101, 01010111) = 3

(10101010, 01010111) = 7

so, $d_{min} = 3$

error can be detected successfully = $(d_{min} - 1)$

$= (3-1) = 2$ bits

error can be corrected upto = $\left(\frac{d_{min}-1}{2}\right) = \frac{(3-1)}{2}$

$= 1$ bit

2)

Data word	Code word
00	11110000
01	00001101
10	10111010
11	01110111

Our received code word = 10101010.

Hamming distance from our code words,

(11110000, 10101010) = 4

(00001101, 10101010) = 5

(10111010, 10101010) = 1

(10101010, 01110111) = 6.

Here, the minimum hamming distance from 10111010 is 1 bit. Hence we can correct the error and retrive the orginal data word from it using Block coding.

Now,

```
11011)10111010101110(110010100
      11011
      -----
       11000
       11011
       -----
         11101
         11011
         -----
          11001
          11011
          -----
             10110
             11011
             -----
              1101 → reminder.
```

As we did not get zero as reminder, we can say we have successfully detected the ~~rror~~ error. ~~done by co~~ So, the error is detected by CRC technique.

02) Given;

Data word $= x^8+x^7+x^5+x^4+x+1$

divisor $= x^4+1$

So, $(x^4)(x^8+x^7+x^5+x^4+x+1)$

$= (x^{12}+x^{11}+x^9+x^8+x^5+x^4)$

Now,

$$x^4+1 \overline{)\, x^{12}+x^{11}+x^9+x^8+x^5+x^4} \,(x^8+x^7+x^5+x^3+1$$

$$\begin{array}{l}
x^{12}+x^8 \\
\hline
x^{11}+x^9+x^5+x^4 \\
x^{11}+x^7 \\
\hline
x^9+x^7+x^5+x^4 \\
x^9+x^5 \\
\hline
x^7+x^4 \\
x^7+x^3 \\
\hline
x^4+x^3 \\
x^4+1 \\
\hline
x^3+1 \rightarrow \text{reminder.}
\end{array}$$

So, code word $= x^{12}+x^{11}+x^9+x^8+x^5+x^4+\underbrace{x^3+1}_{\text{reminder.}}$ is message to be transmitted.

Here,

$$\begin{array}{ccccccccccccc}
x^{12}+ & x^{11}+ & x^{10}+ & x^9+ & x^8+ & x^7+ & x^6+ & x^5+ & x^4+ & x^3+ & x^2+ & x^1+ & x^0 \\
1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1
\end{array}$$

If we corrupt the right-most fifth value, we get, x^4 as the corrupted bit.

Hence, the message,

$$\Rightarrow \underset{1}{x^{12}} + \underset{1}{x^{11}} + \underset{0}{x^{10}} + \underset{1}{x^{9}} + \underset{1}{x^{8}} + \underset{0}{x^{7}} + \underset{0}{x^{6}} + \underset{1}{x^{5}} + \underset{0}{x^{4}} + \underset{1}{x^{3}}$$
$$+ \underset{0}{x^{2}} + \underset{0}{x^{1}} + \underset{1}{x^{0}}$$

$$\Rightarrow x^{12} + x^{11} + x^9 + x^8 + x^5 + x^3 + 1$$

Now,

$$x^4+1 \overline{\big) x^{12} + x^{11} + x^9 + x^8 + x^5 + x^3 + 1} \big(x^8 + x^7 + x^5 + x^3$$
$$x^{12} + x^8$$
$$\overline{x^{11} + x^9 + x^5 + x^3 + 1}$$
$$x^{11} + x^7$$
$$\overline{x^9 + x^7 + x^5 + x^3 + 1}$$
$$x^9 + x^5$$
$$\overline{x^7 + x^3 + 1}$$
$$x^7 + x^3$$
$$\overline{1} \rightarrow \text{reminder.}$$

Here reminder is not zero, Hence error detected.